

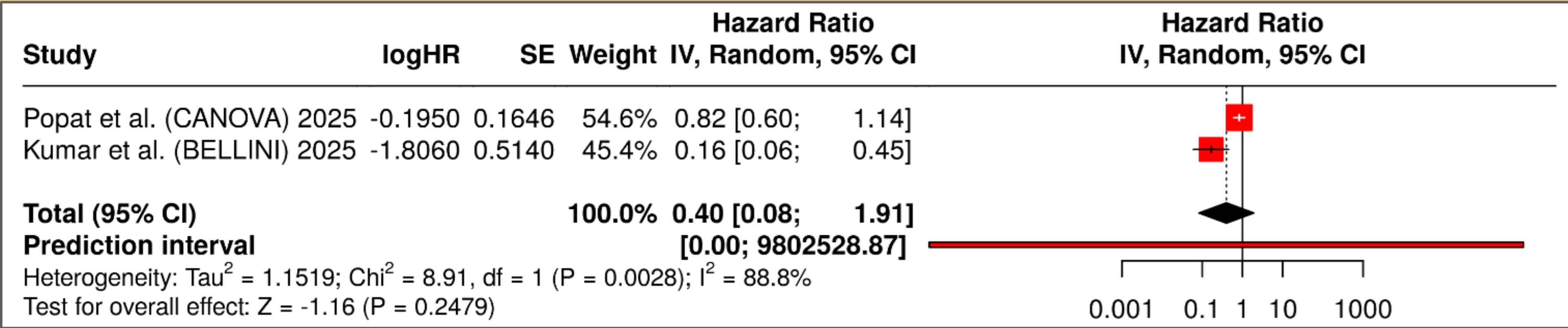
INTRODUCTION

Translocation between chromosomes 11 and 14 [t(11;14)] is one of the most common primary cytogenetic abnormalities in multiple myeloma. Venetoclax is a highly selective, oral BH3 mimetic that binds B-cell lymphoma-2 (BCL-2), inducing apoptosis in BCL-2–dependent plasma cells. Early phase studies suggested promising activity of venetoclax-based regimens, particularly in t(11;14) relapsed/refractory multiple myeloma (RRMM). However, venetoclax has not received FDA approval for the treatment of RRMM, in part due to results from two phase III randomized trials.

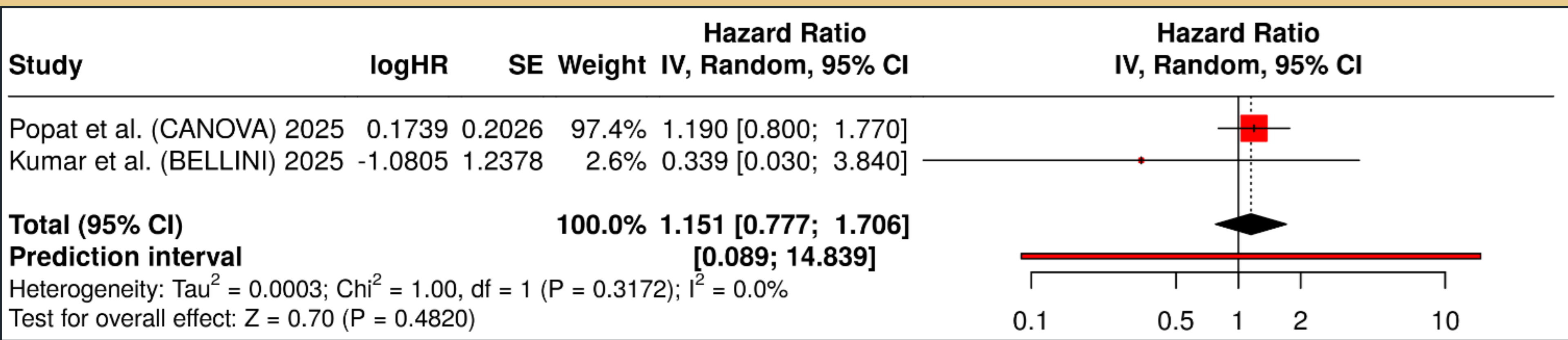
METHODS

A systematic literature search was performed across PubMed-MEDLINE, Embase-OVID, CENTRAL, ClinicalTrials.gov, and Web of Science databases up to January 1, 2025. Randomized phase III trials evaluating venetoclax-based therapy versus standard-of-care regimens in patients with t(11;14)-positive RRMM were included. Progression-free survival (PFS) was the primary outcome. Overall survival (OS) and time to deterioration in disease-related symptoms (TTDDS) were secondary outcomes. Hazard ratios (HRs) were pooled using a random-effects inverse-variance model with heterogeneity assessed using the I² statistic.

Progression-Free Survival (PFS)



Overall Survival (OS)



Time To Deterioration in Disease-Related Symptoms (TTDDS)

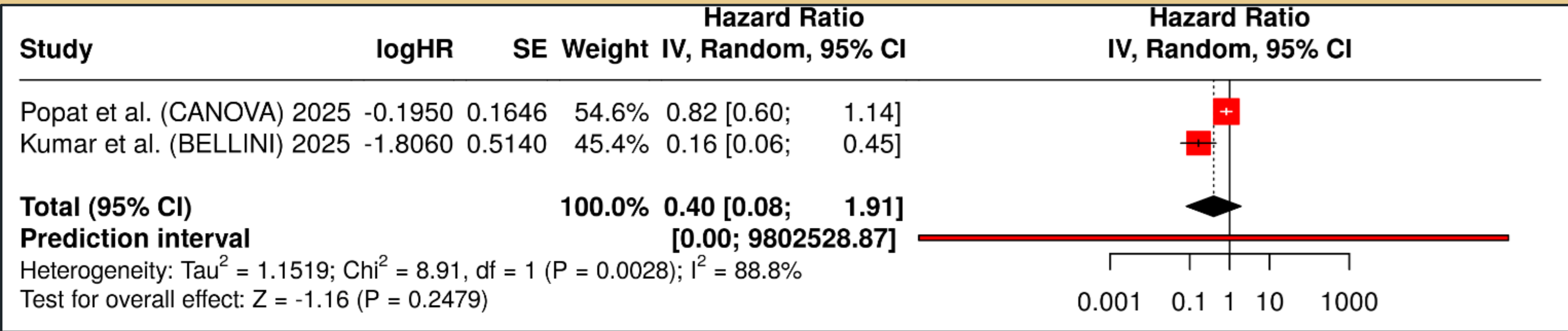


Figure 2 Summary of Pooled Outcomes

RESULTS

A total of 298 patients from two phase III randomized controlled trials (CANOVA and BELLINI) were included. Venetoclax-based therapy was associated with a directionally favorable improvement in PFS (pooled HR 0.40; 95% CI, 0.08–1.91). No pooled OS benefit was observed (HR 1.15; 95% CI, 0.78–1.71). Venetoclax-based regimens demonstrated a trend toward prolonged TTDDS (TTDDS HR 0.70; 95% CI, 0.46–1.05). Substantial heterogeneity was observed for PFS (I² = 88.8%), while heterogeneity was minimal for OS and TTDDS (I² = 0%). None of the pooled efficacy outcomes reached statistical significance.

CONCLUSIONS

Venetoclax-based therapy demonstrates a directionally favorable effect on PFS and TTDDS in t(11;14)-positive RRMM. In BELLINI, despite a small subgroup sample size, venetoclax demonstrated an effect size rarely observed in RRMM, whereas CANOVA was conducted against a highly active comparator. Future trials should prioritize t(11;14) populations and consider non-inferiority or combination-based comparator designs. The lack of statistical significance in pooled outcomes likely reflects differences in trial design and treatment backbone rather than absence of biological activity.

CLINICAL IMPORTANCE

Venetoclax is commonly used off-label in patients with relapsed/refractory multiple myeloma (RRMM) who harbor the t(11;14) translocation, despite lack of FDA approval.

Randomized trial results have been conflicting, leading to uncertainty among clinicians about how effective venetoclax truly is in this setting.

This meta-analysis shows that venetoclax demonstrates a biologically plausible, directionally favorable efficacy signal, but results vary substantially depending on trial design and comparator regimen.

Not all t(11;14) patients derive the same benefit, emphasizing the importance of patient selection, disease context, and treatment backbone.

REFERENCES

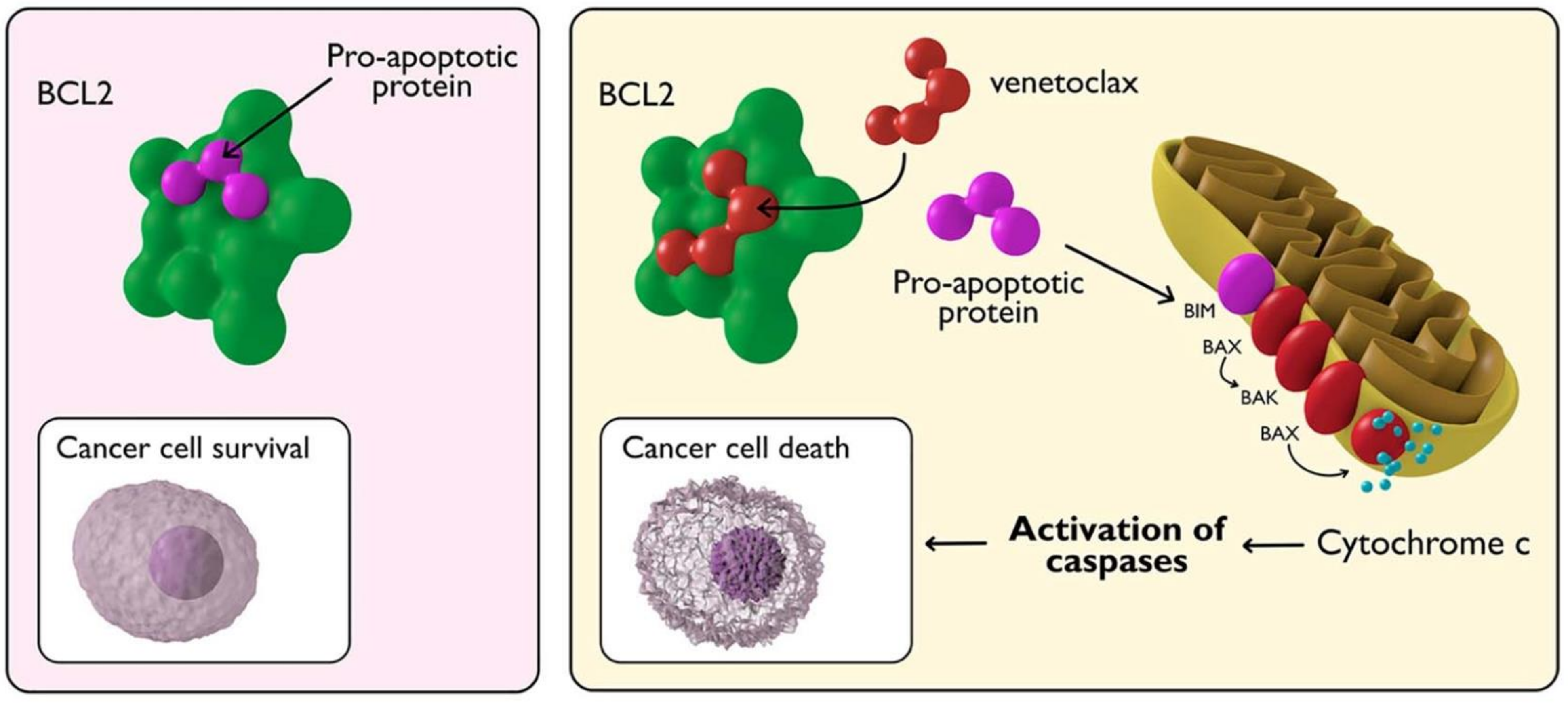


Figure 1 Mechanism of Action of Venetoclax